

As an example, consider the bivariate function,

$$U = f(x_1, x_2).$$

If we let only x_1 to change, with x_2 remaining constant, the rate of change of U with respect to x_1 is called the partial derivative of U with respect to x_1 . Let the change in x_1 be Δx_1 , with x_2 remaining constant. The corresponding change in U is ΔU . Then the difference quotient is given by

$$\frac{\Delta U}{\Delta x_1} = \frac{f(x_1 + \Delta x_1, x_2^0) - f(x_1, x_2^0)}{(x_1 + \Delta x_1) - x_1}$$

If we try to obtain the limiting value of this difference quotient as $\Delta x_1 \rightarrow 0$, we get the partial derivative of U with respect to x_1 . This is denoted by $\frac{\partial U}{\partial x_1}$, $\frac{\partial f}{\partial x_1}$ or simply f_1 .

$$\text{Thus, } f_1 = \lim_{\Delta x_1 \rightarrow 0} \frac{\Delta U}{\Delta x_1} = \lim_{\Delta x_1 \rightarrow 0} \frac{f(x_1 + \Delta x_1, x_2^0) - f(x_1, x_2^0)}{\Delta x_1}.$$

$$\text{Similarly, } f_2 = \lim_{\Delta x_2 \rightarrow 0} \frac{\Delta U}{\Delta x_2} = \lim_{\Delta x_2 \rightarrow 0} \frac{f(x_1^0, x_2 + \Delta x_2) - f(x_1^0, x_2)}{\Delta x_2}$$

Example 11.1: Find the partial derivatives of the function

$$f(x_1, x_2) = x_1^3 + 2x_1^2x_2 + 3x_1x_2 + 4x_2^2$$

To obtain f_1 we have to differentiate the above function with respect to x_1 , treating x_2 as constant.

$$\text{Thus, } f_1 = 3x_1^2 + 4x_1x_2 + 3x_2.$$

Similarly, $f_2 = 2x_1^2 + 3x_1 + 8x_2$. Note that these partial derivatives are functions of the same dependent variables as in case of the original function f .

If $y = f(x_1, x_2)$, then we write $f_1 = f_1(x_1, x_2)$, $f_2 = f_2(x_1, x_2)$.

This is because first derivative at any point was not only dependent on the value of the independent variable, but also on the other variables, which are kept constant. First order partial derivative gives us the marginal values. For example, if the function $U = f(x_1, x_2)$, defines the total utility of the commodities x_1, x_2 , then $\frac{\partial U}{\partial x_i}$, gives us additional utility that can be obtained from the use of one additional unit of x_i . Thus, $\frac{\partial U}{\partial x_i}$ is the marginal utility of x_i (for $i = 1, 2$).

11.2.2 Higher Order Partial Derivatives

We have already seen, if $y = f(x_1, x_2)$, then $f_i(x_1, x_2)$ (for $i = 1, 2$) gives the partial derivatives.

So long as this is true, we can repeat the process of partial differentiation and get higher order partial derivatives. However, once the first order partial derivative

ceases to be a function explicitly of some choice variables as the primitive function, higher order partial derivatives become zero.

Example 11.2:

- i) Differentiate $z = 6x^3 + 5x^2 + 10xy$ partially with respect to x twice.

$$\text{Ans: } z_x = 18x^2 + 10x + 10y; z_{xx} = 36x + 10$$

- ii) Find f_{11}, f_{22} for the function $f(x_1, x_2) = e^{x_1+x_2} + 3x_1 x_2$

Ans:

$$f_1 = e^{x_1+x_2} + 3x_2$$

$$f_2 = e^{x_1+x_2} + 3x_1$$

Thus,

$$f_{11} = e^{x_1+x_2}; 3x_2 \text{ is a constant w.r.t. } x_1 \text{ with derivative as zero}$$

$$f_{22} = e^{x_1+x_2}; 3x_1 \text{ is a constant w.r.t. } x_2 \text{ with derivative as zero}$$

11.2.3 Cross-Partial Derivatives

Consider the function $f(x_1, x_2)$. The first-order partial derivatives of the function are f_1 and f_2 . The cross-partial derivatives are defined as follows:

$$f_{12} = \frac{\partial f_1}{\partial x_2} \text{ and } f_{21} = \frac{\partial f_2}{\partial x_1}$$

By Young's theorem, $f_{12} = f_{21}$.

More generally, for a n - variable function $y = f(x_1, \dots, x_n)$.

$$f_{ji} = f_{ij} \text{ (for all } i, j = 1, 2, \dots, n)$$

Note that such a condition holds when both the partial derivatives exist and one of them is continuous.

Example 11.3:

- i) $f(x, y) = x^3 + y^5 + 4x^2y^4 + 2xy$, find f_{xy} and f_{yx} .

$$f_x = 3x^2 + 8xy^4 + 2y$$

$$f_y = 5y^4 + 16x^2y^3 + 2x$$

$$f_{xy} = 32xy^3 + 2 = f_{yx}$$

- ii) $f(x_1, x_2) = \log(x_1^2 + x_2^2)$. Find the second order partial derivatives.

$$f_1 = \frac{2x_1}{x_1^2 + x_2^2}; f_2 = \frac{2x_2}{x_1^2 + x_2^2}; f_{11} = \frac{2(x_2^2 - x_1^2)}{(x_1^2 + x_2^2)^2}; f_{12} = f_{21} = -\frac{4x_1x_2}{(x_1^2 + x_2^2)^2}; \text{ and}$$

$$f_{22} = \frac{2(x_1^2 - x_2^2)}{(x_1^2 + x_2^2)^2}$$

Check Your Progress 1

- 1) A production function is given by $Q = \frac{1}{2}L^2K^2$, where

Q = level of output;

L = Labour input employed;

K = Capital input employed.

Find the marginal productivities of labour and capital.

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- 2) Given $f(x, y) = x + y$. Find f_{xx} , f_{yy} .

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- 3) Given $f(x, y) = y \sin x^2 + x^3 \cos y$. Find f_{xx} , f_{yy} .

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- 4) Given $f(x, y) = e^{2x+2y}$, prove that $f_{xy} - f_{yx} = 0$.

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11.3 TOTAL DIFFERENTIAL AND TOTAL DERIVATIVE

This section will help you in understanding the difference between total differential and total derivative.

11.3.1 Total Differential

Consider the function $y = f(x_1, x_2)$. By its total differential, we measure the total change in y due to a change in *both* x_1 and x_2 (where x_1, x_2 are assumed to be independent of each other).

However, the change measured by total differential of y is only a linear approximation of the actual change.

The rate of change of y due to a small change in x_1 , keeping x_2 constant is already noted above as f_1 . If the volume of change in x_1 is dx_1 , then the measured change in y due to a change in x_1 will only be equal to $f_1 dx_1$. Similarly, if the volume of change in x_2 is dx_2 , then the measured change in y due to a change in x_2 will be $f_2 dx_2$. Thus, if dy denotes the total change in y when x_1 is changed by dx_1 and x_2 is changed by dx_2 , we can write,

$$dy = f_1 dx_1 + f_2 dx_2.$$

This is called the total differential of the function $y = f(x_1, x_2)$.

Example 11.4: Find the total differential of the function

$$\begin{aligned} y &= ax_1^2 + 2hx_1x_2 + bx_2^2 \\ dy &= 2ax_1dx_1 + 2h(x_1dx_2 + x_2dx_1) + 2bx_2dx_2 \\ &= 2(ax_1 + hx_2)dx_1 + 2(bx_2 + hx_1)dx_2 \end{aligned}$$

11.3.2 Total Derivative

Through total derivative, we measure the rate of change of the dependent variable owing to any change in a variable on which it depends, when none of the variables is assumed to be constant.

Let $y = f(x_1, x_2)$, such that, $x_1 = g(t)$ and $x_2 = h(t)$.

Then we can write

$$\frac{dy}{dt} = \frac{\partial y}{\partial x_1} \cdot \frac{dx_1}{dt} + \frac{\partial y}{\partial x_2} \cdot \frac{dx_2}{dt} = f_1 \frac{dx_1}{dt} + f_2 \frac{dx_2}{dt}$$

This is the total derivative of y with respect to t .

The above result can easily be extended for a n -variable case, namely,

$$y = f(x_1, x_2, \dots, x_n).$$

If $x_1 = x_1(t)$, $x_2 = x_2(t)$, ..., $x_n = x_n(t)$, then

$$\frac{dy}{dt} = f_1 \frac{dx_1}{dt} + f_2 \frac{dx_2}{dt} + \dots + f_n \frac{dx_n}{dt}.$$

Example 11.5:

i) $q = 4x_1 + 3x_2$. If $x_1 = t^3 + t^2 + 1$, $x_2 = t^3 - t^2 - t$.

Find $\frac{dq}{dt}$.

$$\frac{dq}{dt} = f_1 \cdot \frac{dx_1}{dt} + f_2 \cdot \frac{dx_2}{dt};$$

$$\text{where, } f_1 = \frac{\partial q}{\partial x_1}, f_2 = \frac{\partial q}{\partial x_2}.$$

In our example, $f_1 = 4$, $f_2 = 3$.

$$\text{Also, } \frac{dx_1}{dt} = 3t^2 + 2t, \frac{dx_2}{dt} = 3t^2 - 2t - 1.$$

$$\begin{aligned} \text{So, } \frac{dq}{dt} &= 4(3t^2 + 2t) + 3(3t^2 - 2t - 1) \\ &= 21t^2 + 2t - 3 \end{aligned}$$

11.4 DIFFERENTIATION AND ITS APPLICATIONS

This section will cover the process of differentiation of functions in general. Further it will also cover the implicit functions and homogeneous functions. Some applications commonly found in economics are also provided in this section which will help you in concretising the concept of differentiation.

11.4.1 Differentiation of Functions

The formula for total differential can be used to find the derivatives of various types of functions of function (This is also known as **Chain Rule** in the case of total derivatives). Let us consider a few cases.

Case I: Suppose $z = f(u)$, where u is a function of x and y . Here $dz = f'(u).du$.

$$\begin{aligned} \frac{dz}{dx} &= \frac{dz}{du} \cdot \frac{\partial u}{\partial x} = f'(u) \cdot \frac{\partial u}{\partial x} \\ \frac{\partial z}{\partial y} &= \frac{dz}{du} \cdot \frac{\partial u}{\partial y} = f'(u) \cdot \frac{\partial u}{\partial y} \end{aligned}$$

Case II: Suppose, z is a function of x and y which are themselves functions of a single independent variable, t .

$$\text{i) } dz = f(x, y) \text{ where } x = \phi(t), y = \varphi(t). \text{ Then, } dz = \frac{\partial z}{\partial x} \cdot dx + \frac{\partial z}{\partial y} \cdot dy$$

$$\text{But, } \frac{dx}{dt} = \phi'(t) \text{ or, } dx = \phi'(t).dt$$

$$\text{And } \frac{dy}{dt} = \varphi'(t) \text{ or, } dy = \varphi'(t).dt$$

$$\therefore dz = \left\{ \frac{\partial z}{\partial x} \phi'(t) + \frac{\partial z}{\partial y} \varphi'(t) \right\} dt$$

$$\therefore \frac{dz}{dt} = \frac{\partial z}{\partial x} \cdot \phi'(t) + \frac{\partial z}{\partial y} \cdot \varphi'(t) = \frac{\partial z}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial z}{\partial y} \cdot \frac{dy}{dt}$$

Case III: If $z = f(x, y)$ where y is a function of x , then

$$dz = \frac{\partial z}{\partial x} \cdot dx + \frac{\partial z}{\partial y} \cdot dy \quad \text{or, } \frac{dz}{dx} = \frac{\partial z}{\partial x} + \frac{\partial z}{\partial y} \cdot \frac{dy}{dx}$$

Case IV: Suppose z is a function of x and y where x and y are themselves functions of two independent variable u and v . i.e., $z = f(x, y)$ where $x = \phi(u, v)$ and $y = \varphi(u, v)$.

Assume that all functions have known partial derivatives.

Here, $dz = \frac{\partial z}{\partial x}.dx + \frac{\partial z}{\partial y}.dy$, where, $dx = \frac{\partial x}{\partial u}.du + \frac{\partial x}{\partial v}.dv$ and $dy = \frac{\partial y}{\partial u}.du + \frac{\partial y}{\partial v}.dv$

Putting these values in expression for dz we get

$$\begin{aligned} dz &= \frac{\partial z}{\partial x} \left(\frac{\partial x}{\partial u}.du + \frac{\partial x}{\partial v}.dv \right) + \frac{\partial z}{\partial y} \left(\frac{\partial y}{\partial u}.du + \frac{\partial y}{\partial v}.dv \right) \\ &= \left(\frac{\partial z}{\partial x} \cdot \frac{\partial x}{\partial u} + \frac{\partial z}{\partial y} \cdot \frac{\partial y}{\partial u} \right) du + \left(\frac{\partial z}{\partial x} \cdot \frac{\partial x}{\partial v} + \frac{\partial z}{\partial y} \cdot \frac{\partial y}{\partial v} \right) dv. \end{aligned}$$

The partial derivative of z with respect to u is the ratio of dz and du when v is kept constant.

Similarly, the partial derivative of z with respect to v is the ratio of dz and dv when u is kept constant. If u is kept constant, $du = 0$ so that

$$\begin{aligned} \frac{\partial z}{\partial u} &= \frac{\partial z}{\partial x} \cdot \frac{\partial x}{\partial u} + \frac{\partial z}{\partial y} \cdot \frac{\partial y}{\partial u} \\ \text{and } \frac{\partial z}{\partial v} &= \frac{\partial z}{\partial x} \cdot \frac{\partial x}{\partial v} + \frac{\partial z}{\partial y} \cdot \frac{\partial y}{\partial v}. \end{aligned}$$

Rules for finding differentials are similar to those of finding derivatives.

Suppose, u , v and w are functions of x and y . Then,

$$\text{i) } d(u \pm v) = du \pm dv$$

$$\text{ii) } d(uv) = u dv + v du$$

$$\text{iii) } d\left(\frac{u}{v}\right) = \frac{v du - u dv}{v^2}$$

$$\text{iv) } d(\log x) = \frac{dx}{x}$$

$$\text{v) } z = \frac{uv}{w}, \frac{dz}{z} = \frac{du}{u} + \frac{dv}{v} - \frac{dw}{w}$$

11.4.2 Rules Regarding Implicit Functions

In a situation, when there is an implicit functional relation between x and y , such that with a given rise of variation, the values that x and y can take are not independent of each other but are connected in some definite way.

Thus, if the value of x is fixed, the value (or values) that y can take are fixed and not arbitrary. Converse is also true.

The *implicit* function can be expressed in the form $f(x, y) = 0$.

e.g., $ax + by + c = 0$, or, $2x^2 + 3xy + 4y^2 = 0$

Some inverse functions can be expressed in terms of one variable. For example, the first function can be written as

$$x = -\frac{c + by}{a} \text{ or, } y = -\frac{c + ax}{b}.$$

These two *explicit* functions are said to be inverse to each other.

However, all implicit functions may not be expressed in explicit forms where the two variables have been separated. For example, for the second function, viz. $2x^2 + 3xy + 4y^2 = 0$, we cannot express x in terms of y or conversely.

Consider an implicit function $f(x, y) = 0$, where the two variables cannot be separated. Assume that y is a function of x . Here the concept of total differential can be used to find the total derivative $\frac{dy}{dx}$.

The function $f(x, y) = 0$ can be taken as $z = f(x, y)$ where z equals zero.

Now, $dz = f_x \cdot dx + f_y \cdot dy = 0$ (since z is constant). Thus, $\frac{dy}{dx} = -\frac{f_x}{f_y}$.

Example 11.6: $f(x, y) = 2x^2 + 3xy + 4y^2 = 0$. Find $\frac{dy}{dx}$.

Here, $f_x = 4x + 3y$, $f_y = 3x + 8y$. So, $\frac{dy}{dx} = -\frac{4x + 3y}{3x + 8y}$.

Similarly, if it is assumed that x is a function of y , the derivative $\frac{dx}{dy}$ will be equal to $-\frac{f_y}{f_x}$. The derivatives $\frac{dy}{dx}$ and $\frac{dx}{dy}$ are reciprocal to each other, since these two functions are reciprocal to each other.

The second derivative of an implicit function is defined and obtained from the first derivative. If $f(x, y) = 0$ gives y as a function of x , then we have seen that $\frac{dy}{dx} = -\frac{f_y}{f_x}$.

$$\text{Now, } \frac{d^2y}{dx^2} = -\frac{d}{dx} \left(\frac{f_x}{f_y} \right) = -\left\{ \frac{\partial}{\partial x} \left(\frac{f_x}{f_y} \right) + \frac{\partial}{\partial y} \left(\frac{f_x}{f_y} \right) \cdot \frac{dy}{dx} \right\}$$

$$\text{But, } \frac{\partial}{\partial x} \left(\frac{f_x}{f_y} \right) = \frac{f_{xx} \cdot f_y - f_x \cdot f_{yx}}{f_y^2} \text{ and } \frac{\partial}{\partial y} \left(\frac{f_x}{f_y} \right) = \frac{f_{xy} \cdot f_y - f_x \cdot f_{yy}}{f_y^2}. \text{ Substituting these}$$

values and the value of $\frac{dy}{dx}$ we get,

$$\begin{aligned} \frac{d^2y}{dx^2} &= -\left\{ \frac{f_{xx} \cdot f_y - f_x \cdot f_{yx}}{f_y^2} - \frac{f_x}{f_y} \cdot \frac{f_{xy} \cdot f_y - f_x \cdot f_{yy}}{f_y^2} \right\} \\ &= -\frac{1}{f_y^3} (f_{xx} f_y^2 - 2f_{xy} f_x f_y - f_x^2 + f_{yy} f_x^2). \end{aligned} \quad \dots\dots (11.1)$$

Example 11.7 : $f(x, y) = x^2 + xy + y^2$. Find $\frac{d^2 y}{dx^2}$.

$$\frac{d^2 y}{dx^2} = -\frac{2x + y}{x + 2y}$$

$$\frac{d^2 y}{dx^2} = \left\{ \frac{(x + 2y) \frac{d}{dx} (2x + y) - (2x + y) \frac{d}{dx} (x + 2y)}{(x + 2y)^2} \right\} = -\frac{6x^2 + 6xy + 6y^2}{(x + 2y)^3}$$

This can be obtained by putting the values for f_x , f_y , f_{xx} , f_{xy} , f_{yy} in equation (11.1)

11.4.3 Applications in Economics

As you are aware that demand and supply get affected due to multiple factors or, let us say, multiple variables. So, if we wish to find out change in demand function due to the price of the product, then partial derivative is used. Same will apply to the supply function. Few more such concepts, in economics where we use partial derivatives, include identifying of the goods as complementary or substitutes, inferior or normal goods, marginal rate of substitution and marginal rate of technical substitution. Let us look at the applications i) marginal rate of technical substitution and ii) elasticity of substitution, in detail.

i) *Marginal Rate of Technical Substitution (MRTS)*

Let us take the production function as $Y = f(L, K)$

Where Y is the output, L is the labour and K is the capital.

By definition, $MRTS = -(dK/dL)$

Total differential of the production function will be

$$dY = f_L dL + f_K dK$$

where f_L is marginal productivity of labour and f_K is the marginal productivity of capital.

We receive these marginal productivities by calculating the partial derivatives of the production function $Y = f(L, K)$

We know that along a given isoquant there is no change in the output level. So, $dY = 0$.

$$0 = f_L dL + f_K dK$$

Solving the above equation, we get

$$-(dK/dL) = f_L/f_K$$

$$MRTS = f_L/f_K$$

ii) Elasticity of Substitution

Consider the production function $q = f(x_1, x_2)$. The elasticity of substitution measures the ease with which substitution between the two factors can take place.

The degree of substitutability between two factors increases with higher values of the elasticity of substitution.

If it is zero, it will imply impossibility of substitution.

The elasticity of substitution (σ) is defined as the percentage change in the input ratio in response to percentage change in the marginal rate of technical substitution i.e.,

$$\sigma = \frac{\frac{d\left(\frac{x_2}{x_1}\right)}{\left(\frac{x_2}{x_1}\right)}}{\frac{d\left(\frac{f_1}{f_2}\right)}{\left(\frac{f_1}{f_2}\right)}} = \frac{f_1 x_1}{f_2 x_2} \cdot \frac{d\left(\frac{x_2}{x_1}\right)}{d\left(\frac{f_1}{f_2}\right)}$$

$$\text{Now } d\left(\frac{x_2}{x_1}\right) = \frac{x_1 dx_2 - x_2 dx_1}{x_1^2} \quad \text{or, } d\left(\frac{x_2}{x_1}\right) = \frac{x_1 \left(-\frac{f_1}{f_2}\right) dx_1 - x_2 dx_1}{x_1^2} = -\left(\frac{f_1 x_1 + f_2 x_2}{f_2 x_1^2}\right) dx_1$$

$$\left[\because \frac{dx_2}{dx_1} = -\frac{f_1}{f_2} \right]$$

$$\text{Again, } d\left(\frac{f_1}{f_2}\right) = \frac{df_1 \cdot f_2 - df_2 \cdot f_1}{f_2^2} = \frac{(f_{11}x_1 + f_{12}x_2)f_2 - (f_{21}x_1 + f_{22}x_2)f_1}{f_2^2}$$

$$\left[\because df_1 = f_{11}dx_1 + f_{12}dx_2; df_2 = f_{21}dx_1 + f_{22}dx_2 \right]$$

$$= \frac{\left[f_{11}dx_1 + f_{12}\left(-\frac{f_1}{f_2}\right)dx_1 \right] f_2 - \left[f_{21}dx_1 + f_{22}\left(-\frac{f_1}{f_2}\right)dx_1 \right] f_1}{f_2^2}$$

$$= \frac{[f_{11}dx_1 + f_2^2 - f_1 f_2 f_{12} - f_1 f_2 f_{21} + f_{22} f_1^2] dx_1}{f_2^3}$$

$$\text{or, } \sigma = \frac{[f_{11}f_2^2 - 2f_1 f_2 f_{12} + f_1 f_2 f_{21} + f_{22}f_1^2]}{f_2^3} \quad [\text{Using } f_{12} = f_{21}]$$

Thus

$$\sigma = \left(\frac{f_1 x_1}{f_2 x_2}\right) \cdot \frac{(f_1 x_1 + f_2 x_2) dx_1}{f_2 x_1^2} = \frac{f_2^3}{(f_{11}f_2^2 - 2f_1 f_2 f_{12} + f_{22}f_1^2) dx_1}$$

$$= \frac{f_1 f_2 (f_1 x_1 + f_2 x_2)}{x_1 x_2 (2f_1 f_2 f_{12} - f_{11}f_2^2 - f_{22}f_1^2)}$$

The convexity of iso-quant ensures $(2f_1 f_2 f_{12} - f_{11}f_2^2 - f_{22}f_1^2) > 0$ and since $(x_1, x_2, f_1, f_2) > 0$ we have $\sigma > 0$.

The expression of the iso-quant ensures $(2f_1 f_2 f_{12} - f_{11} f_2^2 - f_{22} f_1^2)$ reflects the change of the slope or curvature of an iso-quant. This means that σ is inversely proportional to the degree of convexity of iso-quant. The lower the degree of convexity of the iso-quant, the larger the value of σ and vice versa.

If the two inputs are used only in a fixed proportion, the iso-quant is L-shaped and σ is zero.

In case if the two inputs are perfect substitutes of each other, the iso-quant is a straight line whose MRTS is constant. Here σ is infinity. As σ increases from zero to infinity, substitution between the factors becomes increasingly easier.

Some notes on σ

i) σ varies from one combination of factors to another.

a) $\sigma = \sigma(x_1, x_2)$

b) σ is a lower number

ii) $\sigma_{x_1 x_2} = \sigma_{x_2 x_1}$

iii) If the production function is linear homogeneous, (see below) σ can be written as $\frac{f_1 f_2}{f_{12} \cdot q}$. Here, σ is inversely proportional to the cross-second order partial derivative of the production function.

11.4.4 Homogeneous Functions and Their Properties

A function is said to be homogeneous if multiplication of all its arguments by some arbitrary constant, say k , multiplies the function by k raised to some power.

Thus, the function $f(x_1, x_2)$ is said to be homogeneous of degree n if

$$f(kx_1, kx_2) = k^n f(x_1, x_2).$$

The power of k is called the degree of homogeneity.

For a homogeneous function, the sum of indices for each term of the function is the same.

Thus, the function $f(x_1, x_2) = x_1^2 + x_1 x_2 + x_2^2$ is a homogeneous function of degree two. This is because $(kx_1, kx_2) = k^2 f(x_1, x_2)$.

Similarly, $z = x^3 y + x^2 y^2 + y^4$ is a homogeneous function of degree four.

Property 1: The homogeneous function $z = f(x, y)$ of degree n can be written

$$\text{as } z = x^n \phi\left(\frac{y}{x}\right) \text{ or, } z = y^n \psi\left(\frac{x}{y}\right)$$

You may try to prove from the definition by putting $k = \frac{1}{x}$ or, $k = \frac{1}{y}$.

Property 2: The first order partial derivative for the homogeneous function $z = f(x, y)$ of degree n are homogeneous function of degree $(n - 1)$.

Proof: Let $z = f(x, y)$ be a homogeneous function of degree n . This function can be written as $z = x^n \phi\left(\frac{y}{x}\right)$.

Taking partial differential with respect to x ,